Step 2 - ARM Generic Timer Architecture Overview 1
Reference Note 1
Learning Objectives 1
1. Architecture Overview 1
2. System Counter 2
3. Physical Counter 2
4. Virtual Counter 2
5. Physical Timer 3
6. Virtual Timer 3
7. High-Level Flow 4
8. Physical vs Virtual Time 4
Workflow 5
Knowledge Check 5
Q1 5
Q2 5
Q3 5
Q4 6
Q5 6
Answer Key 6
A1 6
A2 6
A3 6
A4 6
A5 6

Step 2 - ARM Generic Timer Architecture Overview

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture and Cortex-A Generic Timer architecture documentation.

Learning Objectives

After this chapter, you should understand:

- Major components of the ARM Generic Timer architecture
- Relationship between counters and timers
- Physical vs Virtual views of time
- High-level interrupt generation flow

1. Architecture Overview

The ARM Generic Timer Architecture can be viewed as:

ARM Generic Timer Architecture	Description
System Counter	Master source of time shared by all processor cores
↳ Physical Counter (CNTPCT_EL0)	Direct view of system time
↳ ↳ Physical Timer (CNTP_*)	Physical-time based event generation
↳ Virtual Counter (CNTVCT_EL0)	Virtualized view of system time

↳ ↳ Virtual Timer (CNTV_*)	Virtual-time based event generation
Timer Expiry	Compare condition becomes true
Interrupt Generation	Timer signals an interrupt event

Everything starts with the **System Counter**.

2. System Counter

The System Counter is the master source of time.

Responsibilities:

- Measures passage of time
- Shared by all processor cores
- Monotonic (never goes backwards)
- Runs at a fixed rate

Think of it as:

The system's stopwatch.

3. Physical Counter

Provides a direct view of system time.

Example register:

CNTPCT_EL0

Question answered:

What is the current physical time?

4. Virtual Counter

Provides a virtualized view of time.

Example register:

CNTVCT_EL0

Question answered:

What time should a software environment see?

Virtualization details are covered later.

5. Physical Timer

Generates events using the Physical Counter.

Conceptually:

Current Count
↓
Compare Value
↓
Interrupt Event

Uses:

CNTP*

registers.

6. Virtual Timer

Generates events using the Virtual Counter.

Conceptually:

Virtual Count
↓
Compare Value
↓
Interrupt Event

Uses:

CNTV*

registers.

7. High-Level Flow

The Generic Timer works as follows:

System Counter
↓
Current Count
↓
Compare Logic
↓
Timer Expires
↓
Interrupt Event

8. Physical vs Virtual Time

Physical Time
↓
Real hardware time
Virtual Time
↓

Adjusted view of time

Both originate from the same System Counter.

Workflow

System Counter
↓
Physical Counter
↓
Physical Timer
↓
Interrupt

and

System Counter
↓
Virtual Counter
↓
Virtual Timer
↓
Interrupt

The entire ARM Generic Timer architecture is built around these relationships.

Knowledge Check

Q1

What is the foundation of the ARM Generic Timer architecture?

Q2

What is the difference between a Counter and a Timer?

Q3

What is the purpose of a Physical Counter?

Q4

What is the purpose of a Virtual Counter?

Q5

What happens when a Timer reaches its compare value?

Answer Key

A1

System Counter

A2

Counter = Measures Time
Timer = Generates Events

A3

Provides a direct view of system time.

A4

Provides a virtualized view of time.

A5

The timer expires and generates an interrupt event.